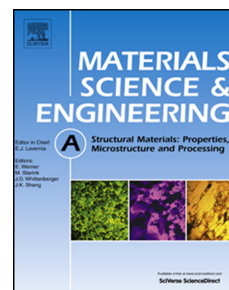


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Effect of Zr, Cr and Sc on the Al-Mg-Si-Mn high-pressure die casting alloys

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Abstract

New aluminium Al5Mg2SiMn casting alloys with Sc, Cr, Zr addition have been developed. The effect of the alloying elements in amount of 0.1 and 0.2 wt.% on the microstructure and mechanical properties of the Al5Mg2SiMn casting alloy produced by the HPDC was investigated. The microstructure was observed by means of the scanning electron microscopy and electron probe microanalysis. The hardness of the aforementioned alloys was determined by Brinell and nanoindentation testing. The microstructure of the HPDC aluminium alloys consists of primarily α -Al phase, the α -Al₁₅(Mn,Fe)₃Si₂ intermetallics and Al-Mg₂Si eutectic cells. Coarse Al₇Cr as well as Al₃Zr intermetallic phases in the alloys with Cr and Zr addition were observed. The most significant improvement in the mechanical properties was achieved in the alloys with Sc addition. Alloying with 0.2 wt.% of Sc exhibits the best combination of strength and ductility with a yield strength of 196 MPa, an ultimate tensile strength of 341 MPa and 10 % of elongation in the as-cast condition. Alloys with Cr and Zr additions present the similar properties in terms of hardness compared to the alloys with Sc addition and the similar tensile properties comparable to the AlMg5Si2Mn (commercially known as “Magsimal-59”) base alloy.

Introduction

New light-weight Al-alloys are important nowadays for energy saving and for the fuel efficiency improvement in the transportation sector. The Al-Mg-Si alloys, relating to the 6xxx series of wrought alloys, are shaped under the so-called soft conditions, usually by extrusion or rolling with the following hardening to their final strength. One of the most important feature of the Al-Mg-Si alloys is the possibility of formation of nanosized precipitates during the heat treatment. The Mg₂Si phase contains 63.2 wt.% Mg and 36.8 wt.% Si, and has a cubic structure with a lattice parameter $a=0.635\text{--}0.640\text{ nm}$ [1]. Thus the mass ratio of Mg to Si is 1.73:1 in Mg₂Si (corresponding to the 2:1 stoichiometry for Mg₂Si). In other words, if the ratio Mg to Si is less than 2.0, then the alloy is designated as possessing an “excess of Si”. Increase of Si content leads to the formation of less stable phases and to the degradation of the mechanical properties in the as-cast state [2], whereas an improvement of the tensile stress and hardness after heat treatment was observed [3–5]. In contrast to

the wrought alloys, in the Al-Mg-Si casting alloys the Mg/Si ratio leads to the concentration of Si in the α -Al matrix, which is not enough for formation of β' and β'' precipitates during aging. Thus there is no need for the heat treatment and maximum values of the mechanical properties can be achieved in the as-cast state. The Mg/Si ratio in casting alloys leads to the Mg_2Si eutectic fraction of around 40 vol.%, which serves as the main hardening phase [6]. Such amount of Mg_2Si eutectic in the Al-Mg-Si casting alloys promotes the improvement in the corrosion resistance, strength and ductility in as-cast state (UTS up to 348 MPa and uniform elongation up to 10%). In this context, further improvement of the mechanical properties can be achieved by alloying.

The Sc addition to the Al-alloys can bring a significant improvement in the mechanical properties. Even a small amount (around 0.2-0.3 wt.% of Sc) can increase the strength of wrought Al-alloys by 100-150 MPa, can improve the weldability and the corrosion resistance [7–9]. Moreover, grain refinement can be achieved by the Sc addition to the Al-alloys. [10–12]. Sc does not form any intermetallic phase with other alloying elements such as Fe, Mg, Mn, Si or Cr [12]. The Al-Si-Sc system can be considered as an exception, where an Al_3Sc phase coexists with the brittle $AlSc_2Si_2$ phase [9,13]. Recently, it was found that the addition of Sc to Al-Si-Mg and Al-Si alloys can change the morphology of the Al-Si eutectic from irregular needle-shaped particles to the rounded and fibrous shape, and that Al grains are refined [10]. It was discussed that the grain refinement mechanism of primary aluminium with Sc is occurred due to the heterogeneous nucleation of α -Al on the Al_3Sc particles [10,12,14]. It was found by Zakharov *et al.* that the strongest effect of the Sc addition on structure and mechanical properties had been observed in Al-Mg based alloys [7]. Several series of industrial wrought alloys with Sc addition have been developed, mostly for the aerospace industry [7,9].

Cr is a common addition to the wrought Al-alloys for grain structure control, grain growth prevention in Al-Mg alloys and for the recrystallization prevention in Al-Mg-Si or Al-Mg-Zn alloys during heat treatment. It can also reduce the stress corrosion sensitivity and improve the material toughness. Moreover, Cr can bond Fe, thereby neutralizing the negative effect of the brittle Fe-rich phases in the casting Al-alloys. But the most effective and commonly used additive for this purpose is Mn [1,15]. Also, Cr, similarly to Mn, forms dispersed hardening particles in a solid solution of aluminum alloys. Leis reported that a recently developed alloy (Maxxalloy-Ultra from SAG Aluminium Lend GmbH&Co), containing 0.10-0.30 wt.% Cr (**Table 1**), possesses in as-cast state one of the highest strength (UTS – up to 350 MPa) among all established Cr-containing casting Al-alloys [16].

Addition of Zr is more popular for the Al-Zn-Mg-Cu alloys[17] and can improve the stress corrosion resistance. It has a grain refinement effect and can improve the strengthening characteristics. It was reported that Zr has a rather positive effect on the hardness and the strength of

the Al-Mg-Si alloys due to the formation of cubic and tetragonal Al_3Zr precipitates in the solid solution [17–19]. Recently, an alloy with a Zr content of 0.1-0.3 wt.% (**Table 1**) has been released (Magsimal-Plus from Rheinfelden Alloys GmbH & Co) [6]. The Mg content is increased in order to avoid the formation of the brittle silicide in the cast structure. In as cast state, the Magsimal-Plus alloy possesses a high strength (UTS – up to 360 MPa) along with the yield stress up to 220 MPa and elongation to failure up to 12% for plates with the a wall thickness of 3 mm [6] (**Table 1**).

Table 1 Nominal composition of commercial Al-Mg-Si-Mn casting alloys and their declared mechanical properties (Al-balance).

Alloy	Chemical composition [wt. %]										Mechanical properties (thickness 3 mm)			
	Si	Fe	Cu	Mn	Mg	Zn	Ti	Zr	Be	Cr	HB	Rp0,2 [MPa]	Rm [MPa]	A [%]
Magsimal	1.8-2.6	0.15	0.05	0.5-0.8	5.0-6.0	0.07	0.2		0.004	-	90-100	160-200	300-350	11-17
Magsimal Plus	2.1-2.6	0.15	0.05	0.5-0.8	6.0-6.4	0.07	0.05	0.1-0.3	0.004	-	No data	200-220	340-360	9-12
Maxxalloy	1.8-2.6	0.2	0.02	0.5-0.8	5.0-6.0	0.07	0.15	-	-	-	85-100	150-195	270-310	10-18
Maxxalloy Ultra	2.2-3.0	0.2	0.02	0.6-0.8	5.6-6.2	0.07	0.15	-	0.004	0.1-0.3	90-105	<220	<350	10-12

In this paper, the effects of Zr, Cr and Sc (up to 0.2 wt.% as the promising additions to Al-Mg-Si-Mn alloys) on microstructure and mechanical properties, have been studied. New alloys with the proposed chemical composition have been casted using the high pressure die casting technique.

Materials and methods

The specimens have been casted at the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Erlangen Germany. The composition of the investigated alloys is shown in **Table 2**. The amount of alloying elements was chosen according to the literature data [6,9,16]. A commercially available software package Thermo-Calc (TC) was used to perform thermodynamic and phase diagram calculations for multi-component systems to investigate the possible effects of the alloying elements on the equilibrium phases in the alloy, using the TCA12:Al-alloys v2.1 database.

Table 2 Nominal composition of the investigated alloys (Al-balance)

Alloy	Chemical composition [wt. %]							Impurity [wt. %]			
	Mg	Si	Mn	Cr	Zr	Sc		Fe	Ti	Be	Cu/Zn
Z1	5.7	2.5	0.7	-	0.1	-		0.2	0.2	0.004	0.15
Z2	5.7	2.5	0.7	-	0.2	-		0.2	0.2	0.004	0.15
C1	5.7	2.5	0.7	0.1	-	-		0.2	0.2	0.004	0.15
C2	5.7	2.5	0.7	0.2	-	-		0.2	0.2	0.004	0.15
S1	5.7	2.5	0.7	-	-	0.1		0.2	0.2	0.004	0.15
S2	5.7	2.5	0.7	-	-	0.2		0.2	0.2	0.004	0.15

AlMg5Si2/M59	5.7	2.5	0.7	-	-	-	0.2	0.2	0.004	0.15
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The commercial Magsimal59® produced by “Rheinfelden Alloys GmbH” (Rheinfelden, Germany) was used as a starting alloy. Blocks of the commercial Magsimal59® were stored in the preheated furnace at 250°C for 24 hours in order to avoid moisture before being transferred into the melt with a temperature of 720°C. For achieving the aimed compositions, commercial AlCr20, AlZr15 and AlSc2 master alloys were successively added into the melt. For degassing the melt was impelled and the density index controlled by a vacuum density tester in combination with the density index balance (MK 3VT and MK3000). The chemical composition was controlled by spark emission spectroscopy (Ametek Spectromaxx). As specimens, plates (**Figure 1**) with the dimensions of 300x150x4 mm were cast using a cold chamber HPDC unit (Frech DAK 450-54 with 4500 kN mold clamping force) at following parameters: melt temperature $T_m=720^\circ\text{C}$, mold temperature $T_d=200^\circ\text{C}$, piston speed $v=2\text{m/s}$, final (post) pressure $p=250\text{ bar}$.

For microstructural examinations, the castings were sectioned, ground and polished to a 0.25 μm finish using the standard metallographic techniques. The final polishing was carried out by vibration polishing using the non-crystallized colloidal silica (0.04 μm).

Microstructural parameters such as the phase average size, its distribution and the volume fraction, the dendritic arm spacing (DAS) as well as the interlamellar space (ILS) of eutectic were investigated and measured using ImageJ software. In order to obtain a statistical average of the distribution, a series of at least 15 micrographs for each specimen were taken. For the calculation of the average grain size at least 70 individual grains were measured. The length of the Al_3Zr phase was calculated for the largest diameter. Owing to the presence of a huge difference in ILS between middle and close to edge parts and the orientation of the lamellas and the way it was cut, the mean ILS value was measured for the mid-central region of the plate only

The microstructural analysis was carried out using a TESCAN Mira3 SEM operating at 15kV and equipped with EDS (Energy Dispersive Spectrometry) detector. The electron probe microanalysis (EPMA) was performed using the JEOL JXA-8100 operating at a voltage of 15 keV, spot size of 10 μm with a step size of 0.25 μm and a dwell time of 500 ms.

The tensile tests at room temperature were performed on the Instron 5967 testing machine at a cross head speed of 1 mm/min, which is equal to an initial strain rate of $8.3 \times 10^{-4} \text{ 1/s}$. The dog-bone shaped tensile specimens with a rectangular cross section with the gauge length of 25 mm and the thickness of 4 mm according to the ASTM E8_E8M_13a were used (**Figure 1**). For each composition 10 tests were performed and the average values of yield strength, ultimate tensile stress and elongation to failure have been calculated.

The hardness was measured by a Brinell tester under a load of 306.56 N (31.25kp) with a ball diameter of 2.5 mm for 10 sec. For each composition 10 tests were performed and the average value has been calculated. The nanoindentations were performed on G200 (Keysight Technologies, USA) using a three-sided Berkovich diamond tip. The continuous stiffness measurement (CSM) technique was used. For each hardness measurement 10 indents were performed with a maximum load of 25 mN. Average hardness values were calculated in a range of 700 nm to the maximum indentation depth of 900 nm in order to avoid surface roughness and indentation size effects.

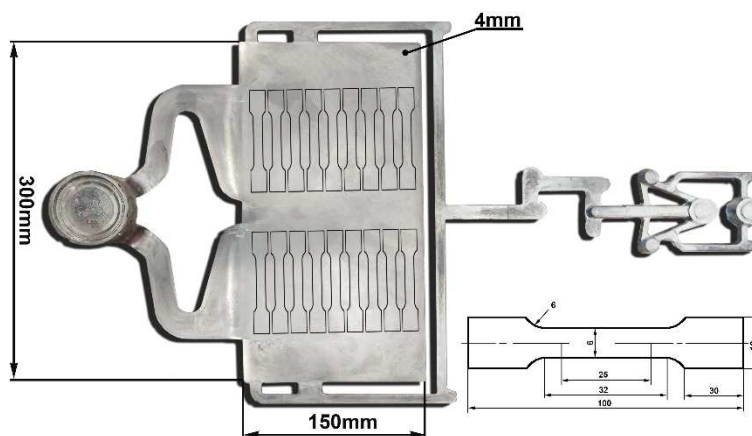


Figure 1 Schematic of the tensile test samples in the as-cast plate with gating system and overflow.

Results

Figure 2 shows the phase diagrams of the proposed chemical composition for the temperature ranges of 450-750°C. In the cast state all studied alloys have new phases in comparison with the base composition. Zr addition leads to the formation of Al_3Zr , Cr to $\text{Al}_{45}\text{Cr}_7$ (more frequently referenced as Al_7Cr [1,20]), and Sc to Al_3Sc phases. Zr has the lowest peritectic concentration among all other alloying elements (0.07 wt.%). Even a high temperature solution treatment of alloys with Zr doesn't lead to the dissolving of a new Al_3Zr phase in α -Al matrix. Cr has the highest solubility in Al (among Zr and Sc) with the peritectic concentration of 0.35 wt.%. New phases in the alloys C1 and C2 can be dissolved during the solution treatment and Cr saturates the α -Al solid solution. The peritectic concentration in Sc-containing alloys amounts to 0.12 wt.% and takes place at 591°C. Therefore in the alloy S1 the Al_3Sc phase can be dissolved during the solution treatment at temperature of 560°C and above.

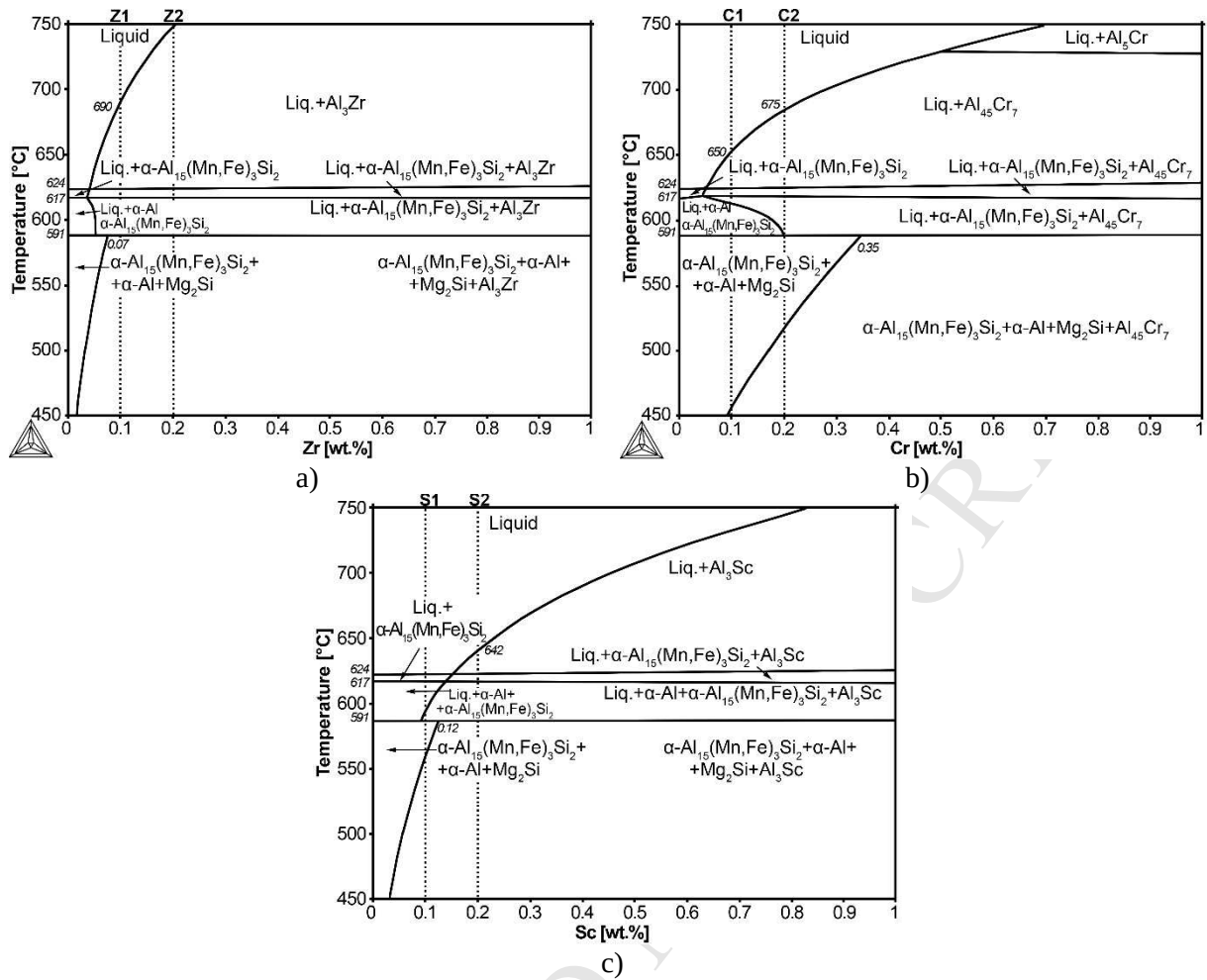


Figure 2 Phase diagrams of the proposed chemical compositions a) $\text{Al}_5\text{Mg}_2\text{SiMn}+\text{Zr}$, b) $\text{Al}_5\text{Mg}_2\text{SiMn}+\text{Cr}$, c) $\text{Al}_5\text{Mg}_2\text{SiMn}+\text{Sc}$.

Microstructural observations

Figure 3a represents the structure of the base alloy ($\text{AlMg}_5\text{Si}_2\text{Mn}$), for which the chemical composition is reported in **Table 1**. The structure of the base alloy is composed of the $\alpha\text{-Al}$ solid solution, $\text{Al}+\text{Mg}_2\text{Si}$ eutectic and the $\alpha\text{-Al}_{15}(\text{Mn,Fe})_3\text{Si}_2$ phase. The $\text{Al}+\text{Mg}_2\text{Si}$ eutectic has a lamellar and/or fibers morphology, where long Mg_2Si plates and fibers alternate with $\alpha\text{-Al}$.

The EPMA maps of the base alloy are shown in **Figure 3 b-c**. Mg as well as Si (**Figure 3 b** and **c**) enrich the dendritic edge and they are mainly accumulated in the eutectic regions. Mn is accumulated mostly in the dendritic edge and forms the Mn-containing phase in the interdendritic space (**Figure 3 d**).

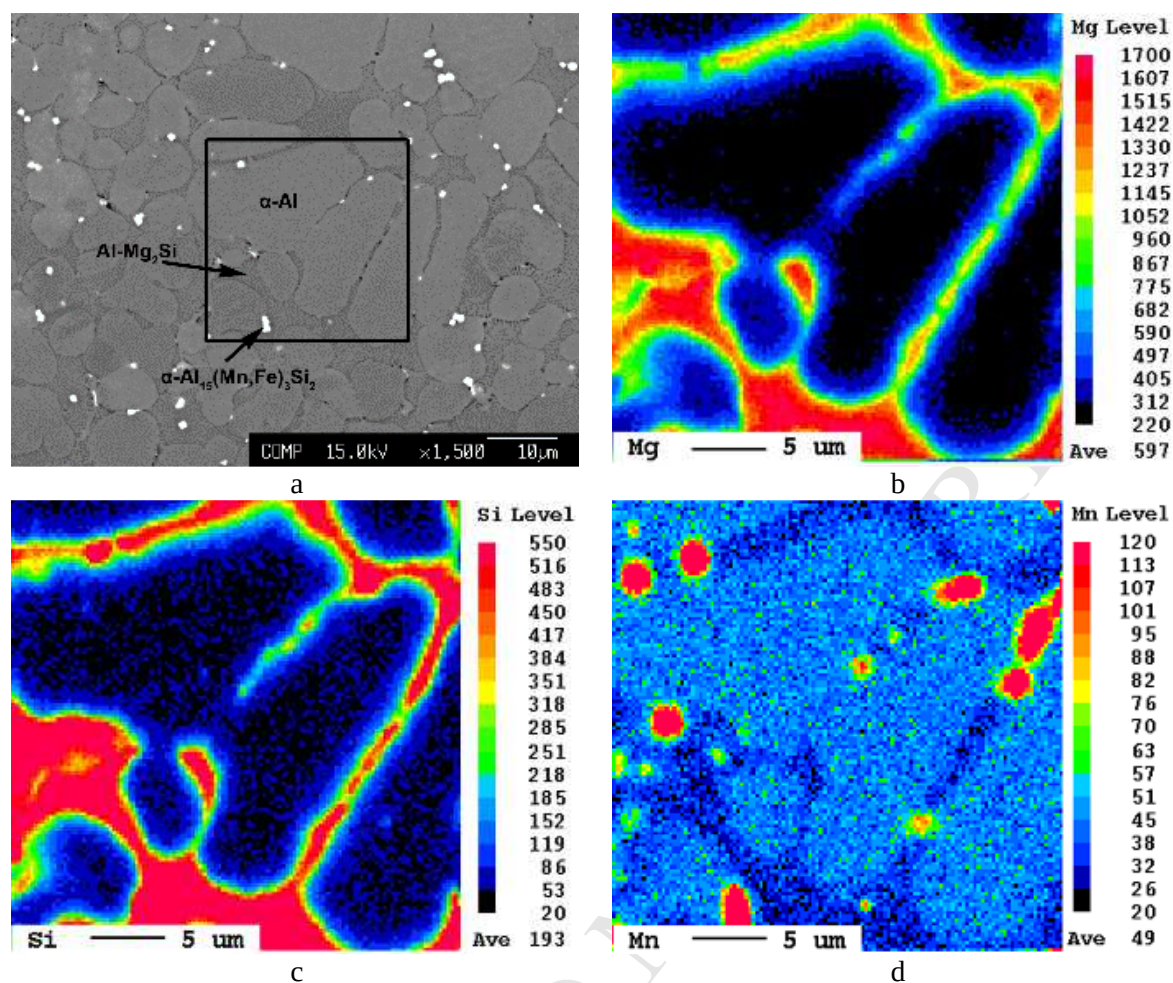


Figure 3 EPMA maps of the M59 alloy a) SEM image of the selected area, and segregation of b) Mg c) Si d) Mn across the selected area

The structures of the studied alloys along with the EMPA maps are given in **Figure 4**, showing the relative x-ray count rate for each element. During solidification, the transition metals (Sc, Zr, Cr) segregate differently to the Al-dendrites [21]. Zr (**Figure 4 a-b**), with the $k \sim 2.5$ [22] is enriched at the aluminum dendrite core and lean at the dendritic edge. It can be seen that the interdendritic area in these alloys is slightly poor with Zr. Cr ($k \sim 2.0$ [22]) as well as Zr also prevail at the dendrite core and lean at the dendritic edge (**Figure 4 c-d**). Over and above a high amount of Cr in the C1 and C2 alloys is presented in the Mn-containing phases. The EMPA map of the alloy with the Sc addition in amount of 0.2 wt.% is presented in **Figure 4 e** and **4 f**. In contrast to Zr and Cr, Sc has a little tendency for the segregation in the Al matrix ($k \sim 1.0$ [22]) and is mainly located in the interdendritic area.

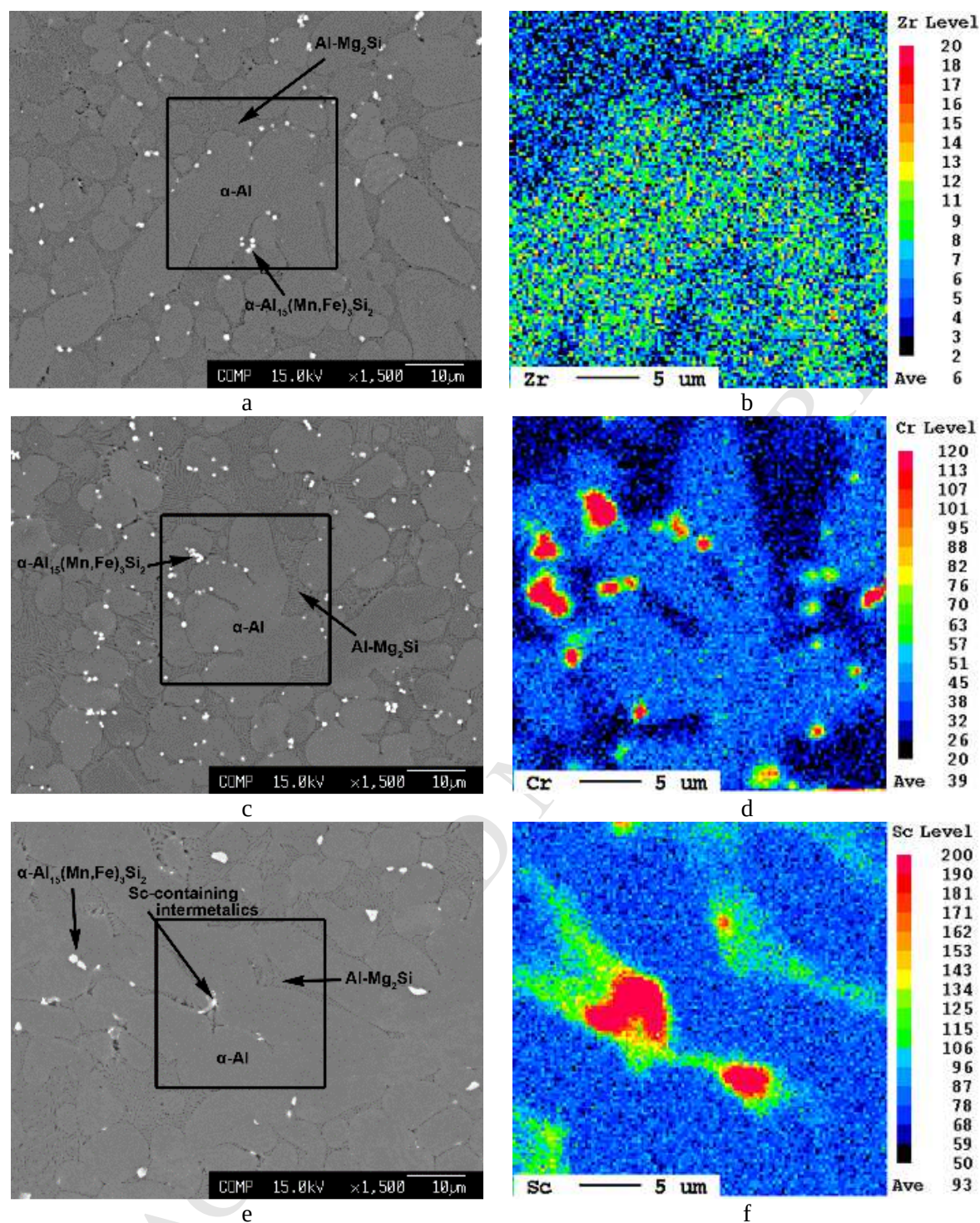


Figure 4 EPMA maps of the distribution of studied alloys with 0.2 wt.% of alloying elements: a-b) Z2, c-d) C2 e-f) S2 alloys

The chemical compositions of α -Al solid solution and intermetallic phases are presented in **Table 3**. Based on the literature data and the measured average compositions, these intermetallic phases can be identified as α -Al₁₅(Mn,Fe)₃Si₂, Al₇Cr and Al₃Zr [1,6,18,20]. In the alloys with Sc addition (**Figure 4** e-f), a small bright phase with the irregular shape is found. Due to the small size

and interposition directly in the immediate region of the Al+Mg₂Si eutectic, a determination of the precise chemical composition by EDS analysis is not possible.

Table 3 Average composition of the α -Al solid solution and the Mn-containing phase in the M59 alloy and Cr-, Zr-phases in the C2 and Z2 alloys [at.%]

Element	α -Al dendrites							Mn-phase							Cr-phase	Zr-phase
	M59	S1	S2	Z1	Z2	C1	C2	M59	S1	S2	Z1	Z2	C1	C2		
Mg	2.8	2.8	2.7	2.8	2.7	2.8	2.8	3.5	3.5	2.4	3.3	3.7	1.3	1.8	1.5	0.8
Al	96.4	95.9	96.3	96.2	96.4	96.1	96.0	75.9	75.9	78.6	78.7	76.9	77.0	75.8	85.2	77.4
Si	0.3	0.4	0.4	0.3	0.4	0.4	0.4	7.3	7.3	6.0	6.5	7.4	7.7	7.0	1.2	0.3
Mn	0.3	0.4	0.3	0.3	0.3	0.2	0.2	10.1	10.1	10.3	10.4	10.4	10.1	10.2	5.5	0.3
Fe	0.1	0.1	0.1	0.1	0.1	0.1	0.1	3.0	2.9	2.4	1.1	1.6	2.5	2.4	0.4	0.2
Cr	-	-	-	-	-	0.1	0.2	-	-	-	-	-	1.4	2.6	5.0	-
Zr	-	-	-	0.2	0.3	-	-	-	-	-	0.1	0.1	-	-	-	20.7
Sc	-	0.1	0.2	-	-	-	-	-	0.1	0.4	-	-	-	-	-	-
Ti	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	1.2	0.3

Figure 5 presents Cr, Zr and Sc containing intermetallic phases, which have been found in C2, Z2 and S2 alloys. The coarse intermetallic phase in the Z2 alloy has an orthogonal like shape with an average length of $13.3 \pm 1.5 \mu\text{m}$ and width of $3.4 \pm 0.8 \mu\text{m}$. In contrast to the Z2 alloy, phases in the C2 alloy have an irregular blocky-like shape and the sizes are inhomogeneous with the minimum length of $6 \mu\text{m}$ up to $27.9 \mu\text{m}$. The Al_3Sc phase has an irregular morphology and is located in the interdendritic space mostly near the $\alpha\text{-Al}_{15}(\text{Mn,Fe})_3\text{Si}_2$ phase and has a similar contrast on the SEM images to the $\alpha\text{-Al}_{15}(\text{Mn,Fe})_3\text{Si}_2$, thus, its identification becomes more complicated (see **Figure 5 c**). The average size of the phases are summarized in **Table 4**.

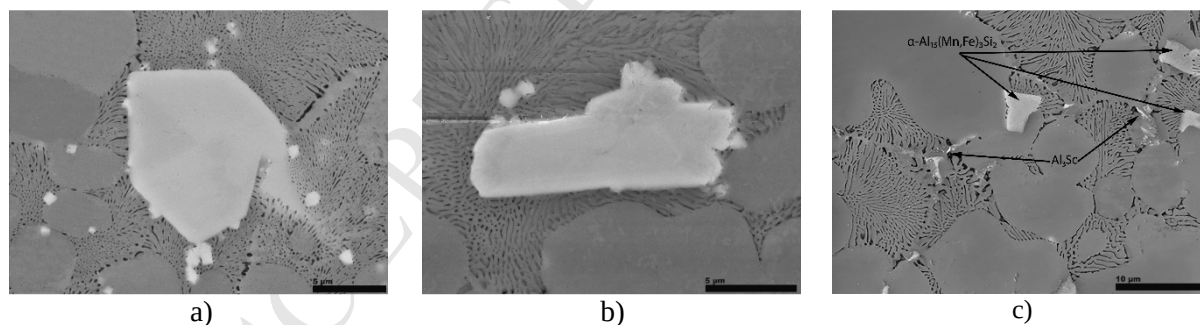


Figure 5 SEM images of the Cr-, Zr- and Sc-containing intermetallic phases in the a)C2 and b)Z2 alloys and c)S2 alloy.

The mean size of the intermetallics and their volume fraction are given in **Table 4**. The volume fraction of the Mn-containing phase as well as the Cr-, Zr- and Sc-containing intermetallics are rather close to those obtained from the thermodynamic calculations for each of the proposed compositions. Nevertheless, the volume fraction of the Al_7Cr , Al_3Zr and Al_3Sc in the alloys with 0.1 wt.% of the alloying element is not measurable due to the low amount and very inhomogeneous distribution across the sample.

Table 4 Volume fraction and mean size of the intermetallic phases in the investigated alloys.

Alloy	Mn-containing phase			Intermetallic phase		
	Size [μm]	V_{meas} [%]	V_{calc} [%]	Size [μm]	V_{meas} [%]	V_{calc} [%]
M59	1.5 $\pm$ 0.2	1.8 $\pm$ 0.2	1.97	-	-	-
				Cr-containing phase		
C1	0.8 $\pm$ 0.4	1.8 $\pm$ 0.37	1.97	*	*	0.38
C2	0.7 $\pm$ 0.3	1.9 $\pm$ 0.35	1.97	13.5 $\pm$ 7.8	0.7 $\pm$ 0.19	0.77
				Zr-containing phase		
Z1	1.3 $\pm$ 0.4	1.8 $\pm$ 0.17	1.97	*	*	0.12
Z2	1.1 $\pm$ 0.4	1.8 $\pm$ 0.19	1.97	13.3 $\pm$ 1.5	0.2 $\pm$ 0.1	0.24
				Sc-containing phase		
S1	1.8 $\pm$ 0.3	1.8 $\pm$ 0.16	1.97	*	*	0.24
S2	1.7 $\pm$ 0.3	1.8 $\pm$ 0.15	1.97	*	0.5 $\pm$ 0.06	0.48

*could not be measured due to the small size and inhomogeneous distribution across the sample.

V_{meas} – results of image analysis; V_{calc} – calculated with ThermoCalc software

Table 5 shows the DAS and the interlamellar space of the eutectic (ILS) of the investigated alloys. After HPDC in as-cast state, α -Al exhibits an equiaxed, globular-rosette morphology. Zr, Cr and Sc additions in such small amount do not affect the grain size. Meanwhile, the ILS decreases with the addition of Cr and Zr. The most prominent effect show alloys with the Sc addition. The ILS decreases almost in two times compared to the M59.

Table 5 DAS and ILS of the eutectic measured for the investigated alloys.

Alloy	DAS [μm]	ILS [μm]
M59	12.3 $\pm$ 7.3	0.42 $\pm$ 0.03
Z1	11.2 $\pm$ 4.2	0.41 $\pm$ 0.02
Z2	11.4 $\pm$ 4.9	0.40 $\pm$ 0.02
C1	11.0 $\pm$ 5.2	0.30 $\pm$ 0.04
C2	10.7 $\pm$ 4.9	0.29 $\pm$ 0.04
S1	11.1 $\pm$ 3.9	0.26 $\pm$ 0.04
S2	11.2 $\pm$ 4.5	0.24 $\pm$ 0.04

Mechanical behaviour in as-cast state

Results from the Brinell hardness measurements as a function of alloy content are presented in **Figure 6 a**. The strongest effect on Brinell hardness was observed for the alloys with Sc addition. The alloy S2 with 0.2 wt% Sc is 14 % harder than the M59, but only 1% harder than the S1 alloy. The average hardness increase for the alloys with 0.1 wt% of Zr and 0.1 wt% of Cr is around 6% as compared to M59. Z2 alloy shows the same hardness value as Z1, whereas the C2 alloy has a comparably high hardness value, similar to that of alloys S1 and S2.

The tensile properties (yield stress, ultimate tensile stress and elongation to failure) are plotted as a function of alloying element and are presented in **Figure 6 b-d**. The results show that the addition of Cr and Zr does not significantly affect the tensile strength, as the maximum increase for the

chromium containing alloys is 5% (**Figure 6c**). The largest improvement in tensile properties is obtained through the addition of Sc (**Figure 6 b-d**). Alloy S1 (with only 0.1 wt% Sc) shows an improvement in strength and elongation by 11% and yield strength up to 16%. An increase of the Sc content up to 0.2 wt.% (S2 alloy) shows increase in the elongation by 9%, in the yield strength as well as in the tensile strength up to 3% in comparison with the S1 alloy. A comparison of the M59 alloy to the S2 alloy shows an improvement in the tensile strength up to 15%, in the yield strength as well as in the elongation by 21%.

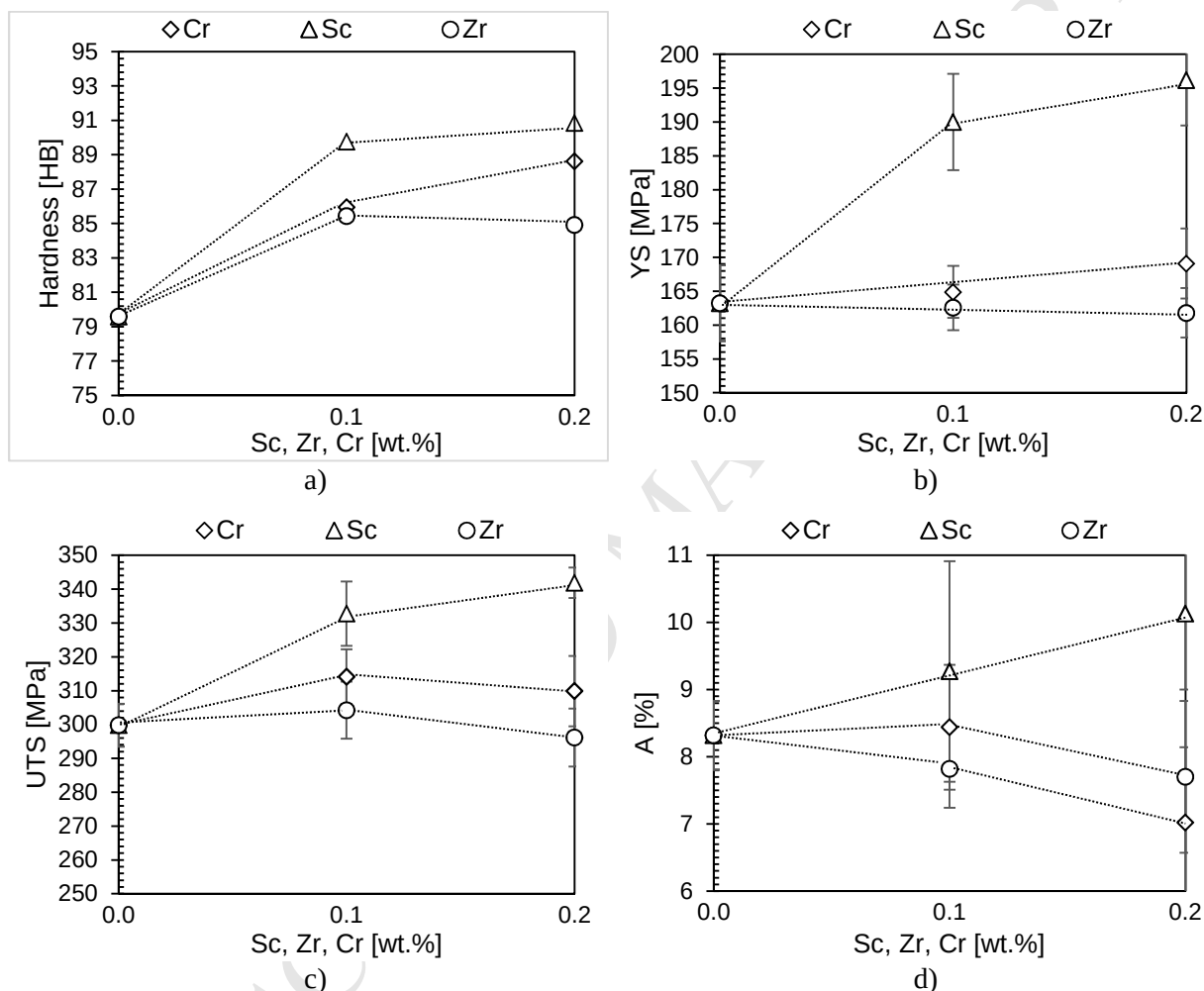


Figure 6 Effect of the alloying elements on the mechanical properties of the studied alloys. a) results of the Brinell hardness tests, b) yield strength c) ultimate tensile stress and d) elongation to failure.

Figure 7 shows the fracture surface in the C2 and Z2 HPDC specimens. On the both of the Al_3Zr and Al_7Cr intermetallics cracks can be seen. However, the fracture occurred in ductile manner.

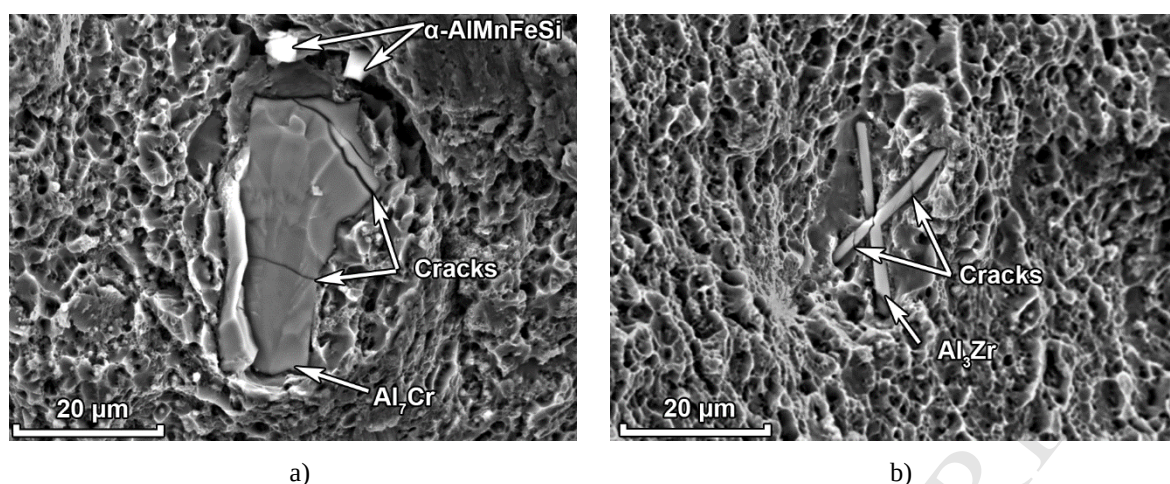
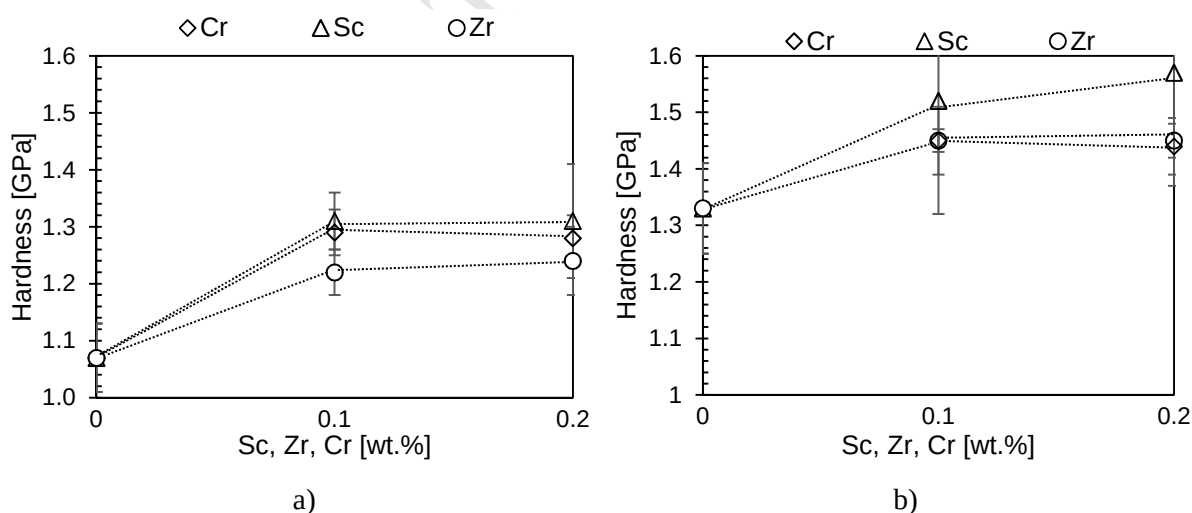


Figure 7 Fracture surfaces of the a) C2 and b) Z2 alloys, show the cracking of the Al_3Zr and Al_7Cr intermetallics.

Using the nanoindentation the local mechanical properties of the α -Al dendrites and Al-Mg₂Si eutectic were measured. As shown in **Figure 8 a**, the hardness values of the dendrites as well as the eutectic (**Figure 8 b**) of the alloys with Zr, Cr and Sc additions are higher in comparison to the M59 alloy. **Figure 8 c-d** shows typical residual impressions in the α -Al dendrites and Al-Mg₂Si eutectic area. The results from the nanohardness measurements show no correlation with the amount of alloying elements within the same group of alloys but rather indicate an equal hardness range for the alloys with the same alloying element. The smallest increase of the α -Al hardness, up to 15%, corresponds to the alloys Z1 and Z2. Alloys with Cr and Sc addition present an increases of up to 20% and 22% respectively. The same trend can be observed in the eutectic, where the highest hardness increase corresponds to the alloys with the Sc addition (**Figure 8 b**).



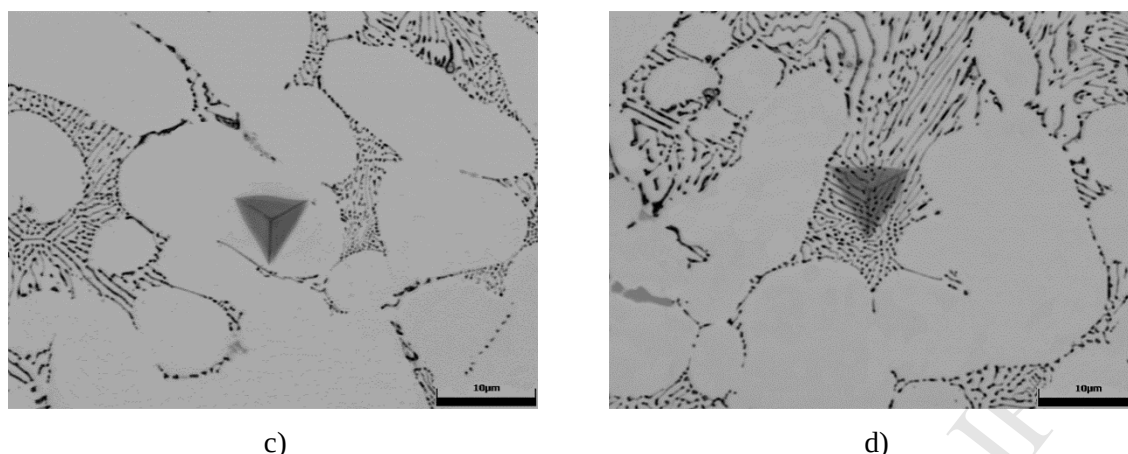
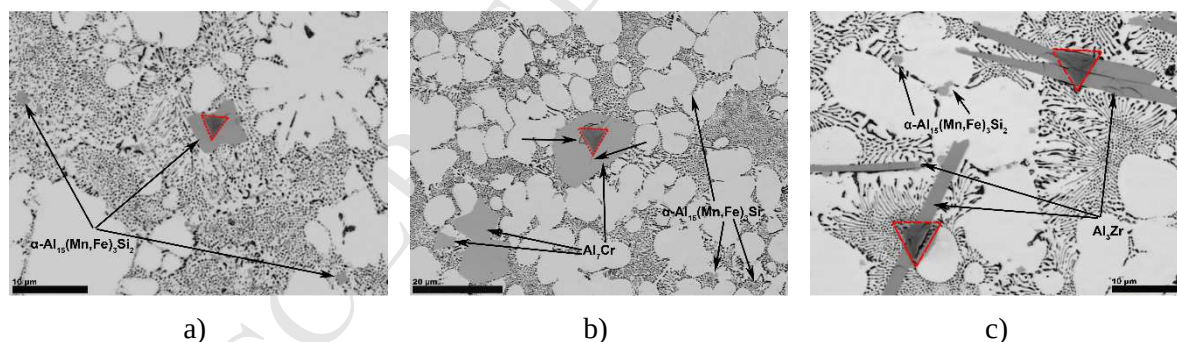


Figure 8 Nanohardness measurements of the a) α -Al matrix and b) eutectic, microstructure of investigated alloy with indent in c) α -Al matrix and d) eutectic as an example.

Furthermore, the indentation behaviour of the α -Al₁₅(Mn,Fe)₃Si₂, Al₃Zr and Al₇Cr intermetallic phases was investigated (SEM: **Figure 9 a-c.**, load-displacement curves **Figure 9 d.** hardness vs. indentation depth **Figure 9 e.**). Since the size of the intermetallic particle is limited, the average hardness value for each intermetallic phase was evaluated at the first hardness plateau at the depth range 100-200 nm. Among investigated intermetallic phases Al₇Cr has the lowest hardness value of 9.57 ± 0.31 GPa, whereas α -Al₁₅(Mn,Fe)₃Si₂ has the highest hardness value of 14.55 ± 0.9 GPa and hardness of Al₃Zr is equal to 10.55 ± 0.1 GPa.



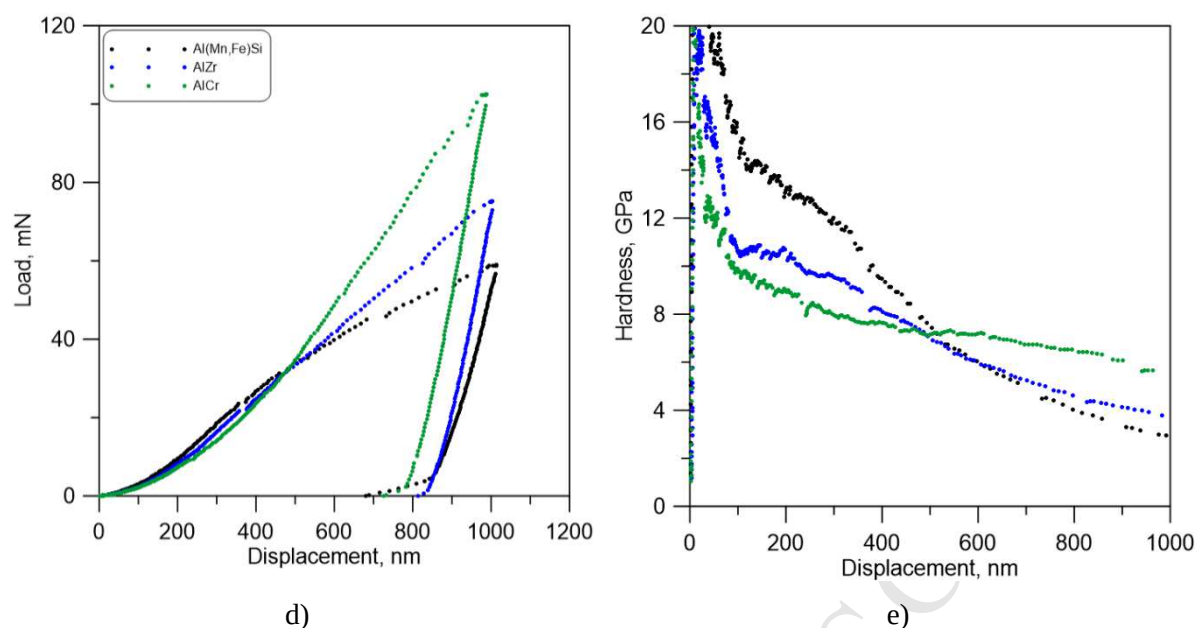


Figure 9 SEM images of a) α -Al₁₅(Mn,Fe)₃Si₂, b) Al₇Cr c) Al₃Zr intermetallic phases, and d) load-displacement, e) hardness-displacement plots.

Discussion

Microstructure

α -Al contains around 0.2 wt. % Si and around 2.7 wt. % of Mg at the eutectic temperature ($\sim 594^{\circ}\text{C}$) [27]. These values are close to those obtained from our analysis at RT (**Table 3**). The addition of Zr, Cr and Sc does not have a significant effect on the distribution of the main elements in the dendrite core (**Table 3**). Due to the different partition coefficients, elements segregate differently. Mg as well as Si (with segregation coefficients in Al less than 1 (~ 0.5 and ~ 0.1 respectively)), enrich at the dendritic edge and mainly accumulate in the eutectic regions (**Figure 3**) [21,22]. The alloys with the Zr, Cr and Sc addition show the formation of Al₇Cr, Al₃Zr and Al₃Sc phases, which are not present in the base alloy. The Al₃Zr intermetallic phases with an orthogonal-like shape were found in the Z2 alloy, but no Al₃Zr phase was observed in the Z1 alloy. This can be due to the low volume fraction of these intermetallic phases (only 0.12% according to the Thermo-Calc, see **Table 4**), an inhomogeneous distribution across a plate and fast solidification during HPDC. Similarly to Zr-series, the coarse Al₇Cr phase was only in C2 alloy observed (**Figure 5 a**). Cr and Zr (segregation coefficients in Al higher than 1) enrich the α -Al dendritic core (**Figure 4 b**). In the C1 and C2 alloys, some amount of Cr was found in the Mn-containing phase (**Table 3** and **Figure 4 d**). Cr is tied up with trace elements, thus forming a stable α -Al_x(Fe,Mn,Cr)_ySi_z (and/or Al₁₂(Cr,Mn)) intermetallics [20,28]. This may lead to the absence of the coarse Cr-containing phase in the C1 alloy. Also, Zr and Sc do not change the chemical composition of the Mn-containing phase. In case of k is close to 1, an element has a little tendency for a segregation in the Al, such as Mn and Sc, and are homogeneously

distributed in the α -Al dendritic core, but mainly it accumulates in the interdendritic area (**Figure 4 e-f**) [29–31]. Apparently, during solidification these elements are pushed to the front of solidification interface. Thus, the Fe,Mn- rich and Sc- rich intermetallics are mainly precipitate in the interdendritic regions. According to literature Al_3Sc intermetallic [1,12] is formed in binary Al-Sc alloys at 0.52 wt. % Sc concentration. For the studied system (**Figure 2 c**) the peritectic reaction occurs at a concentration of ~ 0.12 wt. % Sc and the Sc-rich phase is formed. Microstructural and EMPA investigations confirm the presence of Al_3Sc in the Sc-containing alloys

Mechanical properties

As mentioned earlier, the microstructure of the HPDC Al-Mg-Si alloys consists of the primary α -Al phase, Al- Mg_2Si eutectics and intermetallic phases. It was discussed in literature [18,32–34] that similar concentrations of Cr, Zr and Sc additions have a positive effect on tensile properties and uniform elongation to failure in Al-Mg-Si wrought alloys and Al- Mg_2Si composites, mainly due to grain refinement effect. Notwithstanding, according to **Table 5**, it was found that the addition of Cr, Zr and Sc up to 0.2 wt.% to the Al-Mg-Si casting alloy does not affect the DAS. Probably, for the grain refinement in eutectic casting alloys, higher concentrations of these elements are needed. Therefore, the improvement of the mechanical properties in the investigated alloys cannot be attributed to grain refinement only. On the other hand, a decrease in the ILS with the addition of the alloying elements has been observed (**Table 5**). ILS of the Al- Mg_2Si eutectic structure in the M59 alloy is equal to $0.42\ \mu\text{m}$, which is close to those reported by Shimosaka *et al.*[35]. The addition of Zr does not appear to have a significant effect on the morphology and the mean value of ILS remains in the same range ($0.40\ \mu\text{m}$). However Cr and Sc addition has a more pronounced effect and reduces the ILS up to 0.29 and $0.24\ \mu\text{m}$ respectively.

Sc, Cr and Zr provide similar solution strengthening to the α -Al phase, as can be seen in the nanohardness values in **Figure 8 a**. However the hardness values are not depending on the alloy content. This means, that the lower amount (0.1 wt.%) of alloying elements is sufficient to enrich the solid solution. The excess of alloying elements, which are not solved in the α -Al, forms the Al_3Zr , Al_7Cr and Al_3Sc phases (**Figure 5**). A similar trend is observed for the hardness of the Al- Mg_2Si eutectic cells, which also shows a saturation for the hardness above 0.1 % alloy content for Cr and Zr. The highest values (compared to all studied alloys) of the hardness of the Al- Mg_2Si eutectic cells was observed in the S2 alloys (**Figure 8**). This can be attributed to the segregation of Sc in the eutectic cell, leading there to a reduction of the ILS and the formation of Al_3Sc particles (**Figure 8 a-b**).

Intermetallic phases can act as obstacles to dislocation motion, leading to a strengthening of the alloy. However, if intermetallic particles have a needle shape and/or large size (e.g. Al_xSi silicide in Al-Si alloys, Al_xFe binary or $\text{Al}_x\text{Fe}_y\text{Si}_z$ ternary phases), they can initiate cracks, acting as stress concentrators [2,28,36,37]. Alloying with Mn and Cr can partly reduce this harmful effect, since they

substitute for Fe, and stabilize Fe- and Si-bearing phases, forming fine and compact α - $\text{Al}_x(\text{FeMnCr})_y\text{Si}_z$ intermetallic phase.

Al_7Cr in the C2 alloy forms coarse but rather compact precipitates, which are uniformly distributed, whereas the Al_3Zr phase in the Z2 alloys has a needle-shaped morphology and is rather unevenly distributed. These coarse phases apparently act as a stress concentrator, reducing the yield stress and elongation to fracture of these alloys (**Figure 6 d**). Among the investigated intermetallic phases, only Al_3Sc interacts with the $\text{Al}+\text{Mg}_2\text{Si}$ eutectic seemingly without acting as stress concentrator and has a positive effect on the tensile properties (see **Figure 6 d** and **Figure 8 b**). As shown in **Figure 6 a**, the maximum hardness values was found have Sc-containing and C2 alloys. Yan [38] reported that the impact of Cr addition on the hardness in the simple binary Al-Cr alloys is more pronounced, compared to additions of Zr in Al-Zr alloys. Nevertheless, the Cr and Zr-containing alloys have the lowest elongation to failure. Thus the negative effect of the Al_7Cr and Al_3Zr intermetallic phases is clearly demonstrated (**Figure 6 d**). The Sc containing alloys showed the highest hardness values as well as tensile properties among all studied samples. And, as it was discussed earlier, seemingly this can be a joint effect from the ILS reduction in the Sc-containing alloys, and an interaction of Al_3Sc with eutectic cells.

Conclusions

The segregation of Cr, Zr and Sc in Al-Mg-Si-Mn alloys, local properties of α -Al dendrites and $\text{Al}+\text{Mg}_2\text{Si}$ eutectic as well as the tensile properties have been studied for the new alloys. The Al_7Cr and Al_3Zr intermetallic phases in the alloys with Cr and Zr addition have been detected in addition to the expected and well known phases of the Al-Mg-Si-Mn system. The Al_3Zr phase has an orthogonal shape with an average size of $13.3\pm 1.5\mu\text{m}$, whereas the Al_7Cr phase has an irregular blocky-like shape and sizes from 6 to $28\mu\text{m}$. The Al_3Sc phase with irregular shape has been observed in the interdendritic space of the alloys with Sc addition.

Due to the low solubility of Cr and Zr in Al their addition should be limited in order to prevent formation of the coarse phases and changing of the alloy's microstructure. The addition of Cr in amount of 0.2 wt% (alloy C2) leads to the formation of the Al_7Cr phases. The hardness of the C2 alloy exhibits much higher values due to the presence of the Al_7Cr phase. However, the local hardness measurements of the α -Al dendrites do not show the same trend. The addition of Zr leads to the formation of the Al_3Zr phase but it was observed in the Z2 alloy (0.2 wt.% Zr) only. The hardness of the Z1 and Z2 alloys achieve the same values. Cr and Zr slightly improve the yield strength and ultimate tensile strength, but leads to degradation in elongation to failure. Sc has the most prominent effect on the mechanical properties, even at concentrations of only 0.1 %. Sc acts as a solid solution

strengthening element in the α -Al and segregates to the eutectic cells. There the ILS is strongly reduced in comparison to the base alloy and moreover Al₃Sc precipitates are formed, which are not detrimental to the elongation to failure.

Acknowledgments

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